

Snappy Ubuntu Core for Embedded and IoT Devices

Ubuntu Core is a minimalistic version of Ubuntu. It is lightweight and is designed for use with embedded systems and IoT devices. Snaps are universal Linux packages, which are faster to install, easier to create, and safe to run and work on multiple distributions.



The biggest challenge in delivering Linux applications is dependency resolution. As distributions update underlying libraries frequently, it is not always possible to offer application binaries as offline archives. One can't think of downgrading an application if underlying packages are upgraded, which are in turn required by some more applications. Also, it's not possible to keep multiple versions of the same library in view of different applications. Ubuntu Core has a solution for all this complexity, whereby a single device can host multiple isolated applications with their own dependencies.

Ubuntu 'snaps' are universal packages supported on many Linux distributions and on various devices. Any Linux application can be bundled as a snap, which can be easily deployed across the distributions and across devices with isolated environments. This ensures secure execution and better transactional management with flexible upgradable or downgradable support.

Snaps are based on Ubuntu Core, which is a minimal rendition of Ubuntu, specifically designed for embedded and IoT devices. The core image consists of the kernel, a minimal

set of drivers and a basic file system in the form of a core snap with basic functionality. You can't update the core by adding packages unlike other typical Linux distributions. Each snap is bundled with its own dependencies for the functionality not provided by the core. It supports transactional updates of the snaps, which can be upgraded or downgraded easily without affecting other snaps. Also, snaps can be remotely upgraded.

The snap format file system

When Ubuntu Core is installed, the OS snap gets installed initially. This snap is packed with *rootfs*, providing basic features like network services, standard libraries and daemon management. It is the platform for subsequent snaps offering application logic. This enables universal packaging as all snaps run on the basis of the OS snap provided by the core, eliminating any dependencies with the host environment.

Each snap comes with an isolated file system in *squashfs* format, which gets mounted in a sub-directory under */snap*. For example, when *foo* snap is installed, its file system is mounted under */snap/foo/xx*, where *xx* represents a revision of the snap. There can be multiple